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Studierichting: Informatica
Oefeningenreeks 2D nr. 10.12

Bereken de volgende limiet:

$$\lim_{x \to -\frac{1}{2}} \frac{-6x^2 - x + 1}{-10x^2 - x + 2} = \cancel{\frac{0}{0}}$$

Teller ontbinden:

$$-6x^2 - x + 1 = -(6x^2 + x - 1)$$

$$\Rightarrow -6x^2 - x + 1 = -(3x - 1)(2x + 1)$$

Noemer ontbinden:

$$-10x^2 - x + 2 = -(10x^2 + x - 2)$$

$$\Rightarrow -10x^2 - x + 2 = -(5x - 2)(2x + 1)$$

$$\lim_{x \to -\frac{1}{2}} \frac{\cancel{(2x+1)}(1 - 3x)}{\cancel{(2x+1)}(2 - 5x)}$$

$$= \lim_{x \to -\frac{1}{2}} \frac{1 - 3x}{2 - 5x}$$

Invullen:

$$= \frac{1 - 3\left(-\frac{1}{2}\right)}{2 - 5\left(-\frac{1}{2}\right)} = \frac{5}{9}$$

References